CORE PURE / COMPLEX NUMBERS

De Moivre's Theorem

Questions by topic 13 questions · 115 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

Question 1	*8 marks*	2
Question 2	*5 marks*	3
Question 3	*4 marks*	4
Question 4	*8 marks*	5
Question 5	*10 marks*	6
Question 6	*9 marks*	7
Question 7	*10 marks*	8
Question 8	*9 marks*	9
Question 9	*11 marks*	10
Question 10	*10 marks*	11
Question 11	*12 marks*	12
Question 12	*5 marks*	13
Question 13	*14 marks*	14

1. **(a)** By expanding $(\cos\theta + i\sin\theta)^3$ and using de Moivre's theorem, show that

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$ **[4]**

(b) Hence find all values of θ, where $0 \leq \theta \leq \pi$, that satisfy

$$4\cos^3\theta - 3\cos\theta = \cos 2\theta$$

giving your answers as exact multiples of π. **[4]**

2. Use de Moivre's theorem, by expanding $(\cos\theta + i\sin\theta)^4$ and equating real parts, to express $\cos 4\theta$ in the form $a\cos^4\theta + b\cos^2\theta + c$, where a, b and c are integers to be found. **[5]**

3. Let $z = \cos\theta + i\sin\theta$.
You may assume that, for any positive integer n,

$$z^n - \frac{1}{z^n} = 2i\sin n\theta$$

By considering $\left(z - \frac{1}{z}\right)^3$, find constants A and B such that

$$\sin^3\theta = A\sin\theta + B\sin 3\theta$$ **[4]**

4. **(a)** Use de Moivre's theorem to find the constants A, B and C such that

$$\cos^5\theta \equiv A\cos 5\theta + B\cos 3\theta + C\cos\theta \qquad \textbf{[5]}$$

(b) Hence find the exact value of

$$\int_0^{\frac{\pi}{3}} \cos^5\theta \,\mathrm{d}\theta \qquad \textbf{[3]}$$

5. **(a)** By expanding $(\cos\theta + i\sin\theta)^5$, use de Moivre's theorem to show that

$$\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$$ **[5]**

(b) Hence solve, for $0 \leq \theta \leq \pi$, the equation

$$16\sin^5\theta - 20\sin^3\theta + 5\sin\theta = \sin 2\theta$$

giving your answers as exact multiples of π. **[5]**

6. **(a)** Use de Moivre's theorem to show that

$$\cos^6\theta = a\cos 6\theta + b\cos 4\theta + c\cos 2\theta + d$$

where a, b, c and d are constants to be found. **[5]**

(b) Hence show that

$$\int_0^{\pi/4} \cos^6\theta \,\mathrm{d}\theta = \frac{11}{48} + \frac{5\pi}{64}$$ **[4]**

7. A complex number z has modulus 1 and argument θ.

(a) Show that

$$z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i\sin n\theta, \qquad n \in \mathbb{Z}^+$$ **[3]**

(b) Hence, by expanding $\left(z + \frac{1}{z}\right)^3$ and $\left(z - \frac{1}{z}\right)^3$, show that

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$ **[7]**

8. **(a)** Use de Moivre's theorem to show that

$$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$$ **[5]**

(b) Hence solve the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

giving all roots in the form $2\cos q\pi$, where q is a rational number. **[4]**

9. **(a)** By considering the binomial expansion of $(z - z^{-1})^6$, where $z = \cos\theta + i\sin\theta$, use de Moivre's theorem to show that

$$\sin^6\theta = \frac{1}{32}\big(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta\big) \quad \textbf{[5]}$$

(b) Hence, using the substitution $x = \tan\theta$, find the exact value of

$$\int_0^1 \frac{x^6}{(1+x^2)^4}\,\mathrm{d}x \quad \textbf{[6]}$$

10. **(a)** Use de Moivre's theorem to show that $\cos 6\theta \equiv (1 - 2\sin^2\theta)(16\sin^4\theta - 16\sin^2\theta + 1)$ **[5]**

(b) By solving the equation $\cos 6\theta = 0$, deduce that $\sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$. **[4]**

(c) Hence write down the exact value of $\sin^2\left(\frac{5\pi}{12}\right)$. **[1]**

11. **(i)** Use de Moivre's theorem to show that

$$\cos 5\theta \equiv \cos\theta \left(1 - 12\sin^2\theta + 16\sin^4\theta\right) \quad \textbf{[5]}$$

(ii) Hence determine the range of values of

$$\frac{\cos 5\theta}{\cos\theta}$$

given that $\cos\theta \neq 0$. **[7]**

12. Use de Moivre's theorem to find the constants A, B and C in the identity

$$\sin\theta\cos^4\theta = A\sin\theta + B\sin 3\theta + C\sin 5\theta$$ **[5]**

13. **(a)** Using de Moivre's theorem, show that

$$\sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$$ **[5]**

(b) By using the substitution $\alpha = \left(\frac{\pi}{2} - \theta\right)$, or otherwise, find a similar identity for $\sin^2\theta\cos^4\theta$. **[3]**

(c) Given that $0 < a < \frac{\pi}{2}$ and

$$\int_0^a \left(\sin^4\theta\cos^2\theta + 2\sin^2\theta\cos^4\theta\right)\,\mathrm{d}\theta = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

show that $a = \frac{\pi}{6}$ satisfies this condition and explain why it is the only possible value of a. **[6]**